

ABHAY RAJESH PATLE

AI/ML & Cloud Engineer | Python • AWS • Machine Learning • Automation

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PROFESSIONAL SUMMARY

AI/ML & Cloud Engineer skilled in Python, AWS, machine learning, and automation, with hands-on experience building intelligent systems and scalable cloud applications. Experienced in XGBoost, PyTorch, FastAPI, Generative AI, Amazon Bedrock, AWS Lambda, Docker, and cloud security. Strong foundation in machine learning pipelines, REST APIs, Linux, cloud infrastructure, and MLOps.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, Deep Learning, Computer Vision, XGBoost, Explainable AI (SHAP), Scikit-Learn, Generative AI, RAG, Feature Engineering

Cloud Infrastructure (AWS): AWS Lambda, Amazon Bedrock, IAM, EC2, S3, VPC, RDS, CloudWatch, API Gateway, DynamoDB,

DevOps & Tooling: Docker, Linux Systems Administration, Git, GitHub, CI/CD Pipelines

Backend & Databases: Python, FastAPI, REST APIs, Boto3 SDK, PostgreSQL, SQL,

Frontend & Analytics: React, Streamlit, Power BI, Data Pipelines & ETL Workflows

PROJECTS

NAGPUR PULSE — AI Traffic Risk & Dynamic Police Deployment Support | Lead AI/ML Engineer | Viksit Nagpur Hackathon 2026 (Team of 5)

Technologies: Python, XGBoost, SHAP (Explainable AI), Scikit-Learn, FastAPI, Geospatial Analytics

- Owned and engineered the entire AI/ML pipeline, transforming raw telemetry into structured risk factors using historical accident records, speed variance, traffic volume, and environmental conditions.
- Built and hyperparameter-tuned an optimized multivariate XGBoost model to compute real-time accident risk scores across city road junctions.
- Implemented Explainable AI using TreeSHAP to provide feature-level transparency and interpretable insights directly to law enforcement authorities.
- Designed low-latency asynchronous FastAPI REST endpoints to deliver real-time predictive scores and dynamic police personnel dispatch recommendations.

• **SAHARA — AI-Assisted Mental Healthcare & Wellness Platform** | AI/ML & Full-Stack Engineer | Final Year Project

Technologies: React, Node.js, Express.js, MongoDB, FastAPI, Generative AI, REST APIs, JWT, Pytest

- Developed a full-stack AI-assisted wellness platform using React, Node.js, Express.js, and MongoDB, integrating mood tracking, PHQ-9/GAD-7 assessments, journaling, AI chat, and appointment management.
- Built a FastAPI-based Generative AI service with conversational context, risk-level classification, and automated crisis-risk detection with predefined safety responses and escalation workflows.
- Implemented secure REST APIs using JWT authentication, refresh-token rotation, bcrypt, rate limiting, CORS, input validation, and audit logging, and tested AI workflows with Pytest.

IAM Guard AI Auditor v2.0 | Cloud Security & Serverless Developer | Personal Project

Technologies: AWS Lambda, Amazon Bedrock, IAM, Python, Boto3, Serverless Architecture

- Architected an event-driven serverless cloud security auditor that parses and evaluates IAM policy JSON documents in real time.
- Integrated Amazon Bedrock foundation models to calculate security scores (0–100) and flag over-privileged or hazardous permissions.
- Automated least-privilege remediation templates with side-by-side policy comparisons, reducing cloud configuration vulnerabilities.
- Enforced automated cloud compliance guardrails and Zero-Trust identity controls using Boto3 and AWS Lambda.

AI Plant Identifier & Health Analytics | ML Developer | Personal Project

Technologies: Python, PyTorch, ResNet18, Computer Vision, Streamlit Cloud

- Built an end-to-end deep learning computer vision pipeline using a custom-trained ResNet18 model for plant species and disease identification.

- Implemented Top-5 confidence score analytics and detailed botanical taxonomy classification.
- Developed automated diagnostic PDF report generation and deployed the interactive web application to Streamlit Cloud.

EDUCATION

- Bachelor of Technology (B.Tech) in Computer Science & Engineering | Tulsiramji Gaikwad-Patil College of Engineering & Technology (TGPCET), Nagpur | SGPA: 8.3 | 2023–2027
- Higher Secondary Certificate (HSC) — Science | Late Khushal Kapse Junior College, Gondia | 71.50% | 2022–2023
- Secondary School Certificate (SSC) | Adasi Tanda High School, Adasi | 77.00% | 2020–2021

CERTIFICATIONS & VIRTUAL INTERNSHIPS

- AWS Academy: Cloud Foundations Cloud Architecting
- EduSkills Virtual Internships: AWS AI-Powered Cloud Engineer; AWS Data Engineering Google AI-ML Virtual Internship
- FutureSkills Prime: Fundamentals of Amazon Braket (Quantum Computing) Handling Sensitive Data on the Cloud; Applications of Big Data Analytics (BDA) Popular Software Tools Used in Cloud Computing
- Microsoft, Nirmaan Org & Redington Foundation: 180-Hour AI/ML Intensive Training Program & Hands-on Capstone

COMMUNITY CONTRIBUTIONS & ACHIEVEMENTS

- Viksit Nagpur Hackathon 2026: Finalist/Participant with project Nagpur Pulse (organized by Jan Manthan Foundation, RTMNU & VNIT Nagpur).
- AWS Student Community Day Nagpur 2026 (IIM Nagpur): Official Certificate of Participation; built IAM security solutions inspired by community sessions.
- Google I/O Extended Nagpur 2026 (Infosys MIHAN): Participated in technical tracks on Agentic SRE, Gemini production deployment, and developer challenges.
- AWS Educate Emerging Talent Community (ETC): Active contributing member.